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Lecture due Feb 14, 2024 20:30 IST

Definition 3.1 The **column space** of a matrix **A** is the span of its columns. The notation for it is **CS (A)**. (It is also called the **image** of **A**, and written **Im (A)**.)

Since **CS (A)** is a span, it is a vector space.

Example 3.2 Consider the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$. Its column space is given by

$CS(A) = \text{the span of } \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix}.$

Using that the column vectors are all constant multiples of the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, we find a basis for the column space consisting of the single vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, and find that **CS (A)** is the line $y = 2x$ in $\mathbb{R}^2$.

Example 3.3 Find a basis for the column space of the matrix

$A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 1 & 2 & 3 \\ 0 & 0 & 9 \end{pmatrix}.$

The column space is defined as **Span** $\left(\begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ -3 \\ 3 \\ 9 \end{pmatrix} \right)$. But the second vector is a multiple of the first vector, so it is redundant. Therefore, the column space can be described more simply as the span of the first and third columns:

$CS(A) = \text{Span} \left(\begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ -3 \\ 3 \\ 9 \end{pmatrix} \right).$

These two vectors are linearly independent, so we do need both vectors in the basis for this column space.

Column space concept check I

1 point possible (graded)

What is the column space of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$? Check all that apply.

☐ **CS (A) = 0.**

☐ **CS (A) = Span** $\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$

☐ **CS (A) = Span** $\begin{pmatrix} 1 \\ 0 \end{pmatrix}.$

☐

$$\text{CS}(\mathbf{A}) = \text{Span} \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

☐

$$\text{CS}(\mathbf{A}) = \mathbb{R}^2.$$

☐

$$\text{CS}(\mathbf{A}) = \mathbb{R}^3.$$

☐

$$\text{CS}(\mathbf{A}) = \text{Span} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

☐

$$\text{CS}(\mathbf{A}) = \text{Span} \begin{pmatrix} \sqrt{2}/3 \\ \sqrt{2}/3 \end{pmatrix}.$$

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Column space concept check II

1 point possible (graded)
If you switch two columns of a matrix $\mathbf{A}$, will the new matrix have the same column space?

☐ Yes.

☐ No.

☐ It depends on the matrix.

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Steps to compute a basis for $\text{CS}(\mathbf{A})$:

- 1. Perform Gaussian elimination to convert $\mathbf{A}$ to a matrix $\mathbf{B}$ in row echelon form.
- 2. Identify the pivot columns of $\mathbf{B}$.
- 3. The corresponding columns of $\mathbf{A}$ are a basis for $\text{CS}(\mathbf{A})$.

Proof that the algorithm finds a basis for column space. (*)

Show

In particular,

$$\dim \text{CS}(\mathbf{A}) = \# \text{ pivot columns of } \mathbf{B}.$$

Warning: Usually $\text{CS}(\mathbf{A}) \neq \text{CS}(\mathbf{B})$.

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Previous

Practice with algorithm

Next

1 point possible (graded)

Which of the following sets of vectors is a basis for the column space of the matrix

$$\begin{pmatrix} 1 & 3 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

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☐

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$

☐

$\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix}$

☐

$\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix}$

☐

$\begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix}$

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3. Column space

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